

ESM2. Additional Results

TABLE I: Precision Test Results

Model Configuration	Precision Mean (Std)			
	ResNet50	DenseNet201	Xception	MobileNetV3-Large
Frozen + No Attention (Baseline)	0.6648 (0.0137)	0.6489 (0.0291)	0.3147 (0.0232)	0.3595 (0.0325)
Frozen + Attention	0.7792 (0.0140)	0.7880 (0.0192)	0.4463 (0.0219)	0.4978 (0.0028)
Fine-tuned + Last Layer + No Attention	0.6658 (0.0280)	0.6605 (0.0141)	0.3779 (0.0003)	0.3855 (0.0121)
Fine-tuned + Last Layer + Attention	0.7966 (0.0072)	0.7957 (0.0155)	0.4422 (0.0345)	0.5327 (0.0228)
Fine-tuned + Last 5 Layers + No Attention	0.7371 (0.0177)	0.7090 (0.0335)	0.6328 (0.0146)	0.6366 (0.0147)
Fine-tuned + Last 5 Layers + Attention	0.8472 (0.0085)	0.8557 (0.0113)	0.7195 (0.0157)	0.7064 (0.0089)
Fine-tuned + Last 10 Layers + No Attention	0.8740 (0.0069)	0.8101 (0.0152)	0.8453 (0.0129)	0.6670 (0.0393)
Fine-tuned + Last 10 Layers + Attention	0.8757 (0.0200)	0.8511 (0.0057)	0.8691 (0.0147)	0.7524 (0.0181)
Fine-tuned + Last 25 Layers + No Attention	0.9233 (0.0027)	0.9032 (0.0444)	0.8948 (0.0274)	0.8526 (0.0189)
Fine-tuned + Last 25 Layers + Attention	0.9352 (0.0087)	0.8897 (0.0193)	0.8990 (0.0205)	0.8463 (0.0040)
Fine-tuned + No Attention	0.9348 (0.0232)	0.9561 (0.0175)	0.9727 (0.0039)	0.9595 (0.0094)
Fine-tuned + Attention	0.9448 (0.0190)	0.9730 (0.0033)	0.9697 (0.0041)	0.9703 (0.0051)
From Scratch + No Attention	0.8828 (0.0345)	0.9042 (0.0121)	0.9322 (0.0040)	0.1418 (0.0967)
From Scratch + Attention	0.8848 (0.0310)	0.9030 (0.0231)	0.9438 (0.0015)	0.0056 (0.0000)

TABLE II: Recall Test Results

Model Configuration	Recall Mean (Std)			
	ResNet50	DenseNet201	Xception	MobileNetV3-Large
Frozen + No Attention (Baseline)	0.7387 (0.0155)	0.7128 (0.0236)	0.3645 (0.0251)	0.4479 (0.0453)
Frozen + Attention	0.8146 (0.0183)	0.8199 (0.0257)	0.5017 (0.0236)	0.5850 (0.0068)
Fine-tuned + Last Layer + No Attention	0.7265 (0.0254)	0.7293 (0.0106)	0.4317 (0.0071)	0.4800 (0.0068)
Fine-tuned + Last Layer + Attention	0.8350 (0.0086)	0.8352 (0.0085)	0.5025 (0.0449)	0.6166 (0.0192)
Fine-tuned + Last 5 Layers + No Attention	0.7867 (0.0281)	0.7759 (0.0264)	0.7032 (0.0072)	0.7107 (0.0179)
Fine-tuned + Last 5 Layers + Attention	0.8811 (0.0159)	0.8819 (0.0131)	0.7602 (0.0152)	0.7661 (0.0051)
Fine-tuned + Last 10 Layers + No Attention	0.8984 (0.0104)	0.8400 (0.0207)	0.8781 (0.0151)	0.7339 (0.0428)
Fine-tuned + Last 10 Layers + Attention	0.9025 (0.0093)	0.8753 (0.0029)	0.8977 (0.0099)	0.8129 (0.0166)
Fine-tuned + Last 25 Layers + No Attention	0.9069 (0.0070)	0.8490 (0.0062)	0.8665 (0.0204)	0.8305 (0.0108)
Fine-tuned + Last 25 Layers + Attention	0.9178 (0.0100)	0.8703 (0.0217)	0.8738 (0.0146)	0.8292 (0.0004)
Fine-tuned + No Attention	0.9378 (0.0294)	0.9638 (0.0168)	0.9739 (0.0036)	0.9631 (0.0094)
Fine-tuned + Attention	0.9500 (0.0213)	0.9819 (0.0032)	0.9803 (0.0010)	0.9765 (0.0038)
From Scratch + No Attention	0.9006 (0.0323)	0.9242 (0.0120)	0.9503 (0.0053)	0.1141 (0.0808)
From Scratch + Attention	0.9058 (0.0327)	0.9298 (0.0125)	0.9606 (0.0041)	0.0546 (0.0769)

TABLE III: Loss Test Results

Model Configuration	Test Loss (Mean (Std))			
	ResNet50	DenseNet201	Xception	MobileNetV3-Large
Frozen + No Attention (Baseline)	0.5517 (0.0088)	0.5415 (0.0235)	1.2822 (0.0502)	1.1310 (0.0181)
Frozen + Attention	0.3995 (0.0145)	0.3747 (0.0138)	1.0810 (0.0227)	0.9310 (0.0429)
Fine-tuned + Last Layer + No Attention	0.5230 (0.0072)	0.5379 (0.0225)	1.2121 (0.0051)	1.0789 (0.0143)
Fine-tuned + Last Layer + Attention	0.3998 (0.0178)	0.3735 (0.0155)	1.1140 (0.0109)	0.8551 (0.0403)
Fine-tuned + Last 5 Layers + No Attention	0.4553 (0.0024)	0.4321 (0.0246)	0.7992 (0.0418)	0.6477 (0.0324)
Fine-tuned + Last 5 Layers + Attention	0.3364 (0.0196)	0.3125 (0.0309)	0.7460 (0.0176)	0.5684 (0.0215)
Fine-tuned + Last 10 Layers + No Attention	0.3037 (0.0072)	0.3056 (0.0097)	0.3988 (0.0433)	0.5958 (0.0417)
Fine-tuned + Last 10 Layers + Attention	0.3377 (0.0294)	0.2786 (0.0150)	0.3691 (0.0195)	0.5454 (0.0218)
Fine-tuned + Last 25 Layers + No Attention	0.2779 (0.0063)	0.3743 (0.1145)	0.3443 (0.0130)	0.5018 (0.0452)
Fine-tuned + Last 25 Layers + Attention	0.2489 (0.0288)	0.2784 (0.0264)	0.3497 (0.0163)	0.5051 (0.0342)
Fine-tuned + No Attention	0.1161 (0.0147)	0.0876 (0.0135)	0.0958 (0.0134)	0.1046 (0.0186)
Fine-tuned + Attention	0.1052 (0.0226)	0.0694 (0.0061)	0.0902 (0.0035)	0.0947 (0.0067)
From Scratch + No Attention	0.1597 (0.0132)	0.1455 (0.0056)	0.1119 (0.0093)	2.2607 (1.5662)
From Scratch + Attention	0.1613 (0.0231)	0.1600 (0.0123)	0.1386 (0.0068)	3.5428 (1.6487)